

Problem

In an electronic device, a series circuit is employed that has a resistance of $100\ \Omega$, a capacitive reactance of $5\ \text{k}\Omega$, and an inductive reactance of $300\ \Omega$ when used at $2\ \text{MHz}$. Find the resonant frequency and bandwidth of the circuit.

Step-by-step solution

Step 1 of 1

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi fL}$$

$$C = \frac{1}{2\pi fX_c} = \frac{1}{(2\pi)(2 \times 10^6)(5 \times 10^3)} = \frac{10^{-9}}{20\pi}$$

$$X_L = \omega L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f} = \frac{300}{(2\pi)(2 \times 10^6)} = \frac{3 \times 10^{-4}}{4\pi}$$

$$f_o = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{\frac{3 \times 10^{-4}}{4\pi} \cdot \frac{10^{-9}}{20\pi}}} = 1.826\ \text{MHz}$$

$$B = \frac{R}{L} = (100) \left(\frac{4\pi}{3 \times 10^{-4}} \right) = 4.188 \times 10^6\ \text{rad/s}$$